RESEARCH METHODOLOGY · STATUS REPORT

Approach B Methodology & Latest Findings

Verified physical prefixes, hypothetical target restoration, and the next K3/K4 challenge.

Evidence window P16–P18, through commit 20121eee
Date 01 November 2026
Scope 2D articulated 3R robot · K0–K4 intended
Status Positive development evidence; K3 sparse, K4 unverified

Technical research report / implemented-method addendum
Status date: 1 November 2026
Repository: `Narendhiranv04/latent-causal-tamp` · `approach-b-xy-v1`
Latest independently inspected commit: `20121eee0c337c89134cea621eef7b80d592a008` (P18)
Primary methodological reference: *Approach B: Final methodology — Prefix-wise physical executability and hypothetical target restoration*, 31 October 2026.
Subsequent evidence: Reports `0120–0125`, with this report centered on verified P16–P18 outcomes.
Document status: Companion/addendum to the existing methodology, **not a retrospective replacement of the frozen original PDF**. The proposed P19–P22 work is prospective and has not been executed as of this report.

Executive summary

The central two-outcome Approach B model has been implemented and tested in the original 2D articulated-robot domain. Its purpose is to score externally proposed ordered obstacle relocations by **(X) physical executability** and **(Y) feasibility of the original target manipulation in the hypothetically edited scene**. These are different outcomes: two orders can reach the same edited scene (shared Y), but only one may be executable (different X).

P16 made the development scenes substantially harder by constructing obstacle layouts that analytically close multiple target grasp/placement modes, rather than blocking just one nominal trajectory. P17 then trained matched models with the existing probability factorization and two *documented* enhancements: GEO2 access features and a symmetric within-root counterfactual ranking loss (SYM). Candidate **D: XYQ-GEO-SYM** outperformed matched DIRECT-GEO on the 115 outcome-blind development roots, but its quadratic X/Y coupling was **not shown to produce the preregistered meaningful benefit** over the capacity-matched uncoupled model **E**. P18 froze checkpoints and a K≤2-scoped independent test plan; **no confirmatory scene has been generated and no independent test has been run**.

The requested methodological scope explicitly includes K0–K4, up to six corrective relocations. Presently this remains unfulfilled for **verified K3/K4**: among P16's 87 hard-scene roots, there are only **two verified catalogue-relative K3 repairs and zero verified K4 repairs**. Thus the next work should concentrate on physically realizable K3/K4 development data, matched learning and independent-test readiness, while avoiding new model complexity.

1. The unchanged scientific question and physical protocol

Given an initial 2D tabletop scene `S0`, target `T` and target goal `g_T`, the system receives a finite proposal catalogue of corrective sequences:

```
r = (a1, ..., am),      at = (ot, dt),      0 <= m <= 6.
```

Each `at` moves an obstacle to a specified destination on the **8×8 resting-placement lattice**. The robot is an articulated finite-width planar **3R** arm with continuous joint-space motion, physical grasp/transport/place/release and a finite motion-search budget. The target manipulation is attempted after the selected correction. The learned scorer **does not generate destinations, search for a grasp, synthesize continuous motion, or claim universal impossibility**.

For an ordered corrective prefix `q=r[1:t]`, the binary outcomes are:

- **Xq = 1:** the robot executes *all* proposed relocations in order from the actual achieved state of the preceding relocation (no home reset between moves), **and** connects from its resulting state to the fixed reference configuration `q_ref` with free gripper and no attachment in the edited scene.

- **Ys = 1:** in the hypothetical edited scene `s=S^q`, the teacher can execute the original target task from that **same fixed** `q_ref`, independently of whether the edits could physically be performed.

The terminal X connector makes `Xq=1` and `Ys=1` concatenable into a valid complete repair under the teacher protocol. Alternative orders with identical final poses reuse **one Y** while each has its own X. An attempted fixed-budget failure is recorded as operational 0 with failure metadata; an unattempted/censored outcome is **missing**, not 0.

The four X/Y possibilities distinguish infeasible correction + infeasible target (00), unreachable-but-useful hypothetical rearrangement (01), physically achievable-but-insufficient correction (10), and composable complete repair (11). The no-op case has `X_empty=1` and Y equal to the target-task feasibility in the unedited scene.

2. The implemented F1–F5 methodology

Component	Exact current role	What is learned?
F1: analytic screen	Disallow illegal placement, overlaps, fixed obstacles and other sound static violations before ranking.	**No**
F2: ordered correction	Encode original scene, robot start, ordered object/destination tokens and predecessor ideal scenes; predict physical X coefficient `a_q`.	**Yes**
F3: edited scene	Encode the hypothetical edited scene and target from fixed `q_ref`; predict target Y coefficient `b_s`.	**Yes**
F4: omission	Recompute X/Y for sequences with one proposed action removed, using the **same F2/F3 scorers**.	**No separate F4 head**
F5: XY interaction	Optional learned quadratic coefficient `w_q` between an X query and its parent shared Y query; ablate with `w=0`.	**Yes, but benefit not established**

The original local energy is:

```
E_q(x,y) = -a_q*x - b_s*y - w_q*x*y,       x,y in {0,1}.
```

The **authoritative root-consistent normalized probability** does not treat each local energy as an independently normalized joint when a shared Y has several X children. Instead:

```
P(Y_s = 1)                = sigmoid(b_s)
P(X_q = 1 | Y_s = y)      = sigmoid(a_q + w_q*y)
p_success(q)              = sigmoid(b_s)*sigmoid(a_q + w_q)
p_success(no-op)          = P(Y_0 = 1).
```

For a root containing unique edited-scene keys `S` and ordered-prefix keys `Q`, the joint factorization is:

```
P(y_S, x_Q | root, Q) = PROD_{s in S} P(Y_s=y_s)
                        * PROD_{q in Q} P(X_q=x_q | Y_{s(q)}=y_{s(q)}).
```

The resulting scene marginals are invariant to how many alternative action orders appear in the candidate catalogue. The X/Y interaction is **statistical**, not a discovered physical causal DAG edge. Time/order appears as the ordered F2 input, not as a new latent factor-graph time variable.

3. What changed in the *implemented* model after the original methodology

3.1 GEO2: informative analytic access geometry

P17 augments learned scene coefficients with **GEO2** features computed without reading labels. They include hand-computed clearances and the fraction/availability of target grasp and placement modes in the edited scene. For a corrective action, GEO2 summarizes the moved object's grasp/place access in the **ideal predecessor scene**, including order-sensitive information.

These features are **analytic geometry**, not learned physics or complete motion feasibility proofs. They share calculations with the access-mode closure generator, which raises a legitimate possibility of generator-specific shortcut learning. The matched DIRECT control receives the same GEO2 information, strengthening—but not exhausting—the comparison.

3.2 SYM: symmetric within-root counterfactual supervision

The original specified training loss was:

```
L_B = L_joint + lambda_J * L_travel.
```

P17 candidate D instead optimizes the *implemented extension*:

```
L_D = L_joint + lambda_J * L_travel + kappa * L_SYM.
```

Here `L_joint` is the **exact observed-data root joint NLL** (shared Y deduplicated; unobserved outcomes marginalized); `L_travel` is Huber loss on `log(1+J)` using only complete, independently replayed successful witnesses; `L_SYM` encourages meaningful within-root target-feasibility differences, covering edits that help, edits that hurt and edits that leave feasibility unchanged. P17 freezes `kappa=1`. These additions must be documented in the final methodology; they were **not** part of the original October 31 two-term objective.

For target feasibility logits `b_s` and `b_0`, representative SYM pair losses are:

```
(Y0, Ys) = (0,1):  softplus( -(b_s - b_0) )
(Y0, Ys) = (1,0):  softplus(   b_s - b_0   )
(Y0, Ys) = (0,0) or (1,1): softplus( abs(b_s-b_0)-1 ).
```

The implementation also includes the earlier P14 contrasted edited-scene pairs, with root-level grouping. A ranking auxiliary does **not** automatically improve probability calibration; report both ranking and proper scoring rules.

3.3 Corrective selection and measured travel

The selector screens using F1 and then scores candidate repairs using predicted success and exact/learned costs:

```
C(r) = -log(p_success(r)) + alpha*m(r) + beta*C_d(r) + gamma*J_hat(r).
```

`m` is the exact number of displacements, `C_d` is the exact normalized object-displacement sum, and `J_hat` estimates full robot joint travel including corrective relocations, terminal connector and target manipulation. The rule can also include a development-selected no-op gate. **Travel is not trained from failed paths**. A reduction in travel conditioned on success must not be credited when a model simply avoids necessary repairs.

4. P16 findings: evidence of the intended physical problems

P16 hard-scene result	Measured evidence
Accepted development roots	**87**
Fully labelled X/Y queries	**1,498**, no censored/replay failures
Initially infeasible roots	**69**
K0 / K1 / K2 / K3 / K4, verified minimum in catalogue	**18 / 13 / 11 / 2 / 0**
No verified catalogue repair	**43**
F1-admissible same-endpoint X-discordant order pairs	**23 pairs** across 20 roots
Order pairs with certified failing order	**7**
Harmful-edit contrast roots (Y0=1, Yedit=0)	**12**
Connector-phase X negative cases	**20** (budget-relative)
P16 CPU use	**16,832 / 24,000 CPU-s**

P16 replaced nominal-sweep obstruction with **analytic grasp/place mode closure**. The old P15 generator usually failed to block alternative target grasps. P16 used finite-mode access maps and intended blocker sets, and verified whole-scene physical outcomes with the unchanged teacher. One fully witnessed three-move example has Y0=Y1=Y2=0 and Y3=1; the reverse order has a certified X failure. This is a valid **physical example** of the intended methodology, not statistical proof that the learned model solves K3 generally.

Critical K caveat. `K_intent` is a construction request. `K_geom` is the analytic access-based intention/bound under its declared assumptions. `K_catalogue` is the fewest moves of a *verified complete candidate* in the registered catalogue. Neither `K_intent=4` nor planner-budget failure of shorter alternatives proves global four-move necessity. P16 observed `K_geom 4 → verified K4` in **0 of 3** tested roots. Forty-three roots had no complete witnessed repair, often due to bounded motion planning.

5. P17 findings: learning and ablations on outcome-blind development roots

The primary outcome-blind development population combines P15 pilot (28 roots) and P16 V2.1 (87 roots): **115 roots total**, **76 initially infeasible**, **31 with a verified candidate repair**, and **39 initially feasible**. Five family-grouped folds × three seeds × six arms produced **90 completed runs**, at **38,138 / 48,000 CPU-s** including a transparently recorded runner restart/fix.

Method / same development setting	Verified repair success, infeasible roots	Unnecessary repair, feasible roots	Notes
XYQ-BASE (B)	5.7%	19.7%	Original XY model without GEO2/SYM
DIRECT-GEO (A)	14.9%	22.2%	Matched geometric inputs
XYQ-GEO-CON (C)	23.2%	15.4%	Earlier asymmetric contrast auxiliary
XYQ-GEO-SYM (D)	**27.2%**	**12.0%**	Predeclared P17 primary candidate
XY-UNC-SA-GEO-SYM (E)	22.4%	**9.4%**	Capacity-matched uncoupled challenger
XYQ-GEO-SYM-ORDINV (G)	19.3%	13.7%	Order-invariant F2 control
Analytic-access reference	15.8%	0.0%	Nonlearned trade-off, not direct capacity-matched
Verify-first ceiling	40.8%	—	Evaluator-only catalogue availability

Preregistered comparisons (paired root/family bootstrap, development only):

- **D vs B:** +0.215 corrective success (95% CI [+0.136,+0.298]); unnecessary repairs −0.077. Development gate met.
- **D vs DIRECT-GEO:** +0.123 (95% CI [+0.061,+0.189]); unnecessary repairs −0.103 (95% CI [−0.179,−0.034]). Development DIRECT-superiority gate met.
- **D vs E (coupling):** +0.048 corrective success, **below** the frozen +0.10 meaningful-benefit threshold; joint NLL is not better. **Quadratic coupling is not supported** as an essential ingredient; E is the simpler practical challenger.
- **D vs order-invariant G:** order-concordance gain +0.18 (95% CI [+0.047,+0.307]), but strict exact-permutation and certified-only attribution deserves an additional read-only diagnostic. No claim of robust long-horizon ordering from these few pairs alone.
- Harmful edited-scene (1→0) sign accuracy improved from about **0.23 (B) to 0.90 (D)**; edited-Y AUROC improved from about **0.68 to 0.94**. These are within-development metrics.

Ablation interpretation. D's superiority over matched DIRECT establishes a whole-method-and-training advantage for the tested setup. It does **not isolate** the X/Y factorization alone: D additionally uses explicit X/Y labels, SYM and travel training, and the selection rules are chosen separately on inner folds. GEO2 is responsible for a substantial part of the improvement, so generator-specific shortcuts remain a threat.

K3/K4 issue. Across the outcome-blind development roots, only two K3 examples and **no K4** support model evaluation. Even if a model successfully chooses one K3 witness, that is an anecdote rather than reliable stratum-level performance.

6. P18 freeze and its current limitations

P18 froze six arms as **15-checkpoint ensembles** (five folds × three seeds), taking the mean candidate success probability and predicted travel, then selecting under one pooled development-chosen rule. This **differs from P17's out-of-fold three-seed predictions with fold-specific rules**. P17's headline 27.2% must not be presented as measured independent performance of the frozen P18 ensemble.

P18 created a **300-construction-slot** confirmatory *plan* under the reserved `bxyg21confirm` namespace; **zero** confirmatory roots have been generated. The plan extrapolates about 226 accepted roots, 84 repairable infeasible roots and 41 K≥2 roots. These estimates derive from relatively small development mechanism strata and do not establish credible K3/K4 test power. The originally reported ±0.107 “worst-case” half-width is calculated as for a single Bernoulli proportion, whereas the paired model comparison uses differences in `{-1,0,1}` and needs its own paired variance/power analysis.

The checkpoint bytes are local, not committed; the published hash list alone is not sufficient for long-term reproducibility if those local files are lost. Preserve an independently accessible, hash-checked backup before any independent collection.

The original 300-slot plan remains immutable but **is not an adequate final K3/K4 confirmation plan**. The next work must use versioned development and, ultimately, a separately frozen and independently generated K3/K4 cohort.

7. Current implementation-versus-proof status

Method claim	As of P18	Remaining proof
X/Y physically composable teacher protocol	Implemented with replay-verified witnesses	More long-prefix witnesses and external physical generalization
Root-consistent normalized X/Y model	Mathematically implemented, tests/ replays pass	Independent predictive and selection evaluation
Useful versus harmful edits	Strong improvement in development	Independent transfer and calibration
Executable action order	Some true order twins; development improvement	Strict certified/permutation diagnostics and enough K3/K4 order roots
Quadratic X/Y term	Implemented and fairly ablated	No meaningful benefit demonstrated; optional in deployment
Cost-aware ranking	Implemented and measured	Success–cost trade-off on longer repairs
K3 (three necessary corrections)	**2 catalogue-relative verified examples**	Substantially more distinct witnessed roots, held-out performance
K4 (four necessary corrections)	**No verified example**	First physically executable 4-move witnesses, candidate-relative shorter-action audit, held-out performance
Independent 2D confirmation	Frozen proposal only	Separate authorization, one-time test, no post-test changes
3D / hardware	Outside completed evidence	Separate future stage; not implied by present data

8. Proposed next implementation: P19–P22, *without new architecture*

- **P19: preflight and methodology alignment.** Verify P17/P18 metrics, source/custody and checkpoint retention, order diagnostic, K_geom assumptions and paired power. Record GEO2/SYM in the methodology addendum. No teacher or training.
- **P20: K3/K4 development scene construction and physical witnesses.** Attempt analytically valid *backward-feasible* 3-/4-blocker arrangements and distinct mode closures, complete prefix/omission query designs, exact true-order twins, neutral/harmful controls. Freeze each catalogue before teacher results. Collect bounded development evidence with 3 sequential, versioned waves and truthful failures. Target ≥20 distinct verified K3 and ≥20 K4 if attainable; limited-data thresholds do not justify strong statistical claims.
- **P21: matched retraining and K-stratified cross-validation.** Same D/E/DIRECT/B/G (and C if budget permits), fixed model architecture and X/Y semantics, family/geometry-grouped OOF evaluation. Report unconditional versus availability-conditioned repair success, unnecessary repair, X/Y probabilities, strict ordering, omission, K3/K4, travel and uncertainty.
- **P22: methodology/evidence freeze and new independent-test plan.** Preserve P18; state whether K3 and K4 are physically verified and learned. Freeze candidate and challenger checkpoints and one-shot analysis. Do **not** generate an independent cohort until a distinct authorization is given.

The detailed implementation contract is supplied separately in `BXY_P19_P22_K3_K4_IMPLEMENTATION_INSTRUCTIONS.md`.

Why a no-result outcome can still be important

The frozen 2D 3R embodiment, 8×8 grid, limited parking cells, grasp-mode family and fixed teacher search budgets may make many analytically designed K4 tasks operationally unreachable. If P20 cannot find complete K4 witnesses within honest bounded effort, the correct outcome is to identify whether the binding limitation is construction geometry, relocation, connector, target skill or candidate catalogue. A failed attempt does **not** license declaring physical impossibility, fabricating K4, changing labels, increasing teacher budgets only for hard examples, or rewriting previous claims.

9. Provenance and references

Authoritative methodology (historical): `Approach_B_Final_Methodology_2026-10-31.pdf` (supplied in this project).
P16 report: `0123_BXY_P16_METHODOLOGY_FIDELITY_AND_HARD_SCENES.md`.
P17 report: `0124_BXY_P17_TARGETED_LEARNING.md`.
P18 report: `0125_BXY_P18_METHOD_FINAL_READINESS.md`.
Implementation: `bxy_p16.py`, `bxy_p17.py`, `bxy_p17_data.py`, `bxy_p17_eval.py`, `bxy_p18.py` at the same frozen commit.
Nature of evidence: repository report and code inspection; recorded tests and logged physical witnesses are not the same as an independent rerun of every experiment. No P19–P22 output or independent test result is implied by this document.

Bottom line: The current 2D Approach B model and physical X/Y protocol are credible development implementations. P17 provides positive matched development evidence for geometry-informed counterfactual repair selection, while coupling remains optional. **The agreed K0–K4 scope is not complete until K3/K4 exist in sufficient replay-verified numbers and are assessed out of fold and independently.** The planned P19–P22 work should pursue exactly that—without complicating the model.